

Supplementary Information

Unconcentrated markets, captive buyers: incumbency in Türkiye’s public IT procurement

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S1 Data construction

S1.1 Source and IT scope

Award records were collected from EKAP (ekapv2.kik.gov.tr) by searching domain keywords. Search matches substrings of tender titles, which admits unrelated tenders. The keyword *erp* retrieved 618 contracts, of which 546 are not IT (it matches, among others, the place name Yaverpaşa and *tuz serpici*, a salt spreader). The phrase *teknik destek* retrieved 1,186 contracts, of which 798 are hospital or university staffing and cleaning contracts. An earlier count-based cleaning had removed 394 obvious false positives from 13,418 records, leaving 13,024.

We re-audited this set by value. All 150 contracts with the highest value were classified by hand (108 IT, 1 call-centre service, 19 mixed, 22 not IT). For the remainder, ordered title rules assign each contract to one of four scope classes: IT, IT-enabled call-centre services, ambiguous, and not IT. Works contracts are ambiguous only if they install an IT or operational-technology system, such as city security cameras, SCADA, fibre, or a data centre, and are otherwise not IT. Staffing contracts for data entry are ambiguous; bundled cleaning or catering contracts without an IT system are not

IT. Random samples from every rule were inspected, and seven rule outcomes for top contracts were overridden by hand. Table S1 reports the result. The main sample consists of the IT and call-centre classes.

EKAP’s award list has one row per tender identifier, so tenders with several lots appear with a single winner. We found 116 main-sample titles that suggest multiple lots and no framework-agreement call-offs (the word *çerçeve* appears only in the sense of “within the framework of”).

Table S1: Scope classes after the value-based audit (13,024 contracts that passed the earlier count-based cleaning).

Scope class	Contracts	Nominal value (bn TRY)	Share of value (%)
IT	9,981	25.64	56.5
IT-enabled call-centre services	10	7.62	16.8
Ambiguous (excluded from main sample)	1,406	4.12	9.1
Not IT (removed)	1,627	8.03	17.7
Total	13,024	45.42	100.0

S1.2 Firms, buyers, and markets

Firms. Names were canonicalized by upper-casing with Turkish locale, removing punctuation and whitespace, and stripping legal-form suffixes from the end of the name, subject to a minimum remaining length. Joint ventures were kept separate from their members. The merge groups of the roughly 300 largest firms were reviewed by hand. Seventy-two brand groups were confirmed, including firms that changed legal form (for example from limited to joint-stock company) or name while keeping the same brand in non-overlapping years. Three uncertain brand pairs were left unmerged. The process reduces 4,788 recorded names to 4,389 firms. By a name-pattern heuristic, 536 of these firms (302 in the main sample) are natural persons. In the released data, every winner that is a natural person or lacks an explicit legal-entity form (690 firms) is replaced by a stable pseudonym.

Buyers. Thirty-six generic buyer labels pool units from two or more provinces, among them the provincial health directorates and hospital unions of the Ministry of Health. We split them by the province recorded for each tender (1,001 of the 13,024 contracts; 34 labels and 732 contracts in the main sample), which turns 3,453 labels into 3,663 buyer nodes. Six national bodies with offices in two provinces were not split. The recorded labels are used as a sensitivity check throughout.

Buyer sectors. Buyers were assigned to 11 sectors by ordered keyword rules on the buyer name. Relative to an earlier coding, 1,423 contracts changed sector, mainly because upper-casing without Turkish locale had prevented mixed-case names from matching, and because several rules matched unintended substrings.

Product markets. Contracts were assigned to 14 product markets by ordered rules on the tender title, most specific first. In the main sample, 97% of contracts receive a specific market. To validate the rules, a random sample of 300 main-sample titles (seed 7) was classified independently, without access to the rule labels. These reference labels were produced with AI assistance (Claude) and have not been checked by human coders. Agreement is 77.0% (Cohen’s $\kappa = 0.74$); excluding 45 titles flagged as genuinely ambiguous, it is 82.4% ($\kappa = 0.80$). Health information systems has precision and recall of 0.99. Among classes with at least 10 reference titles, the weakest are custom software, maintenance, and education technology (F1 0.45–0.56), and the residual “other IT” class is not reliable. Of the 69 disagreements, 22 are clear rule errors, 16 reflect a convention choice between maintenance and the product maintained, 27 are multi-product bundles, and 4 are barely IT.

Dates, procedures, values. Dates are tender dates joined by tender identifier (coverage 100%, October 2010 to March 2026). Procedures are open, restricted, and the negotiated procedures of Article 21(a)–(f); the extract contains no direct procurement under Article 22. Values are deflated to 2025 lira with the annual average of the consumer price index (2003=100; 2026 from the linked

monthly series). With the domestic producer price index (Yİ-ÜFE) instead, count-based indices are unchanged. Seven single-variant value classifications move across the 1,000 or 1,800 boundary, and no robust classification changes.

S1.3 Descriptive statistics by buyer sector

Table S2: Buyer sectors (main sample). Value in bn 2025 TRY; procedure shares in % of contracts.

Buyer sector	Contracts	Firms	Buyers	Value	21(b)	21(f)	Open
Health	2,950	554	1,098	30.93	28.5	9.4	61.8
Municipal / local	2,688	1,084	667	19.93	2.2	44.1	53.4
Other public	1,295	646	351	10.88	2.5	41.1	54.7
Education	1,054	519	330	4.68	5.1	31.8	62.6
Infrastructure / transport	644	337	123	7.72	2.8	39.8	57.0
Agriculture / environment	395	231	98	4.29	2.3	31.4	63.5
Social / finance	303	156	42	5.20	3.6	39.3	55.4
Provincial administration	223	125	74	0.99	1.3	25.1	62.8
Security / interior	202	142	64	1.50	3.5	26.2	66.8
Defence	191	96	29	1.89	1.0	3.7	95.3
Justice	46	32	14	1.25	26.1	28.3	43.5
All	9,991	2,814	2,890	89.26	10.5	29.6	59.0

S1.4 Article 21 of Law No. 4734

Translations are ours. The chapeau reads: “The negotiated procedure may be used in the following cases.”

- (a) No tender was submitted in an open or restricted procedure.
- (b) Before 25 May 2018: the tender must be held urgently because of sudden and unexpected events such as natural disasters, epidemics, or a danger of loss of life or property, or of events the contracting authority could not foresee. Law No. 7144 (Art. 11, in force 25 May 2018) inserted: “or in cases that have special features in terms of construction technique, or in which the contracting authority determines that the work must be carried out urgently to ensure structural safety or the safety of life and property”.
- (c) The tender must be held urgently because of special defence- or security-related circumstances.
- (d) The procurement requires research and development and is not subject to serial production.
- (e) The technical and financial characteristics cannot be specified with the necessary precision because the goods, services, or works are original and complex.
- (f) Purchases of goods, materials, or services whose estimated cost is below a monetary limit, updated every 1 February (for example 157,923 TRY in 2014 and 3,406,508 TRY in 2026).

In cases (b), (c), and (f), no notice needs to be published; at least three firms are invited to submit qualification documents and price offers. In our data all 2,957 main-sample 21(f) contracts are goods or services, and from February 2014 on, 99.96% are below the limit in force.

S2 Concentration

Table S3 reports supplier concentration by product market. Bootstrap intervals are bias-shifted percentile intervals (1,000 resamples of contracts within market). “No max” removes the market’s

largest contract; “winsor.” caps contract values at the 99th percentile of the main sample. Table S4 reports medians of HHIs computed within calendar years for market-years with at least 30 contracts. The bias-corrected count index is the unbiased estimator $10,000 \sum_i n_i(n_i - 1)/[n(n - 1)]$, where n_i is firm i ’s number of contracts and n is the market-year total.

Table S3: Supplier concentration by product market (main sample; HHI on 0–10,000).

Product market	Contracts	Firms	HHI count [95% CI]	HHI value [95% CI]	Value, no max	Value, winsor.	CR4 value (%)
Health information systems	2,189	205	431 [405, 458]	831 [692, 985]	868	811	53
ERP / management software	1,515	599	84 [71, 97]	132 [73, 219]	120	96	16
Custom software / web / mobile	1,117	734	25 [22, 29]	264 [41, 744]	176	110	27
Software licences	1,047	375	103 [89, 118]	275 [188, 441]	276	270	25
Network / data-centre infrastructure	748	390	72 [60, 87]	281 [151, 523]	234	226	28
Computers / peripherals	696	467	40 [34, 47]	170 [69, 363]	146	155	18
Maintenance / support services	626	297	82 [69, 98]	301 [196, 481]	303	296	25
GIS / city information	553	203	353 [273, 444]	408 [228, 842]	385	347	33
Cybersecurity	390	194	193 [143, 263]	476 [193, 1,002]	442	340	37
Physical security / surveillance	385	222	84 [69, 104]	212 [123, 373]	208	210	20
Other IT	277	215	79 [58, 111]	767 [0, 2,275]	603	296	46
Education technology	205	128	225 [147, 346]	1,062 [196, 2,936]	752	537	54
Smart city / traffic / OT	169	106	171 [125, 234]	2,580 [503, 5,917]	823	568	66
Call centre / help desk	74	46	690 [363, 1,160]	8,908 [3,534, 10,000]	1,685	1,391	98
All markets pooled	9,991	2,814	39 [36, 41]	149 [40, 504]	92	87	18

Table S4: Median within-year supplier HHI by product market.

Product market	Years	Count HHI	Count HHI, bias-corr.	Value HHI	Value HHI, no max
Health information systems	16	671	597	1,091	1,081
GIS / city information	12	650	369	1,503	1,206
Physical security / surveillance	4	427	108	1,098	926
Cybersecurity	7	416	156	1,666	1,263
Maintenance / support services	14	343	91	1,627	1,244
Software licences	15	325	187	1,029	868
Network / data-centre infrastructure	15	274	68	1,038	937
Computers / peripherals	15	265	44	903	762
ERP / management software	15	226	119	688	521
Custom software / web / mobile	15	176	37	743	601
All markets pooled	17	69	55	256	209

Decomposition. The pooled index satisfies $H_{agg} = \sum_m w_m^2 H_m + C$, where w_m is market m ’s share and $C = \sum_i \sum_{m \neq m'} s_{im} s_{im'} w_m w_{m'}$ collects the terms of firms active in several markets. By count, $H_{agg} = 38.6$, $\sum_m w_m^2 H_m = 26.7$ and $C = 11.9$; $\sum_m w_m^2 = 0.117$ and the demand-weighted mean of market HHIs is 179. By real value the corresponding figures are 148.6, 127.9 and 20.7, with a weighted mean of 1,278 (534 without the largest contract). A quarter of firms are active in more than one market; they hold 69% of contracts.

S3 Incumbency and null models

S3.1 Subgroups and sensitivities

Definitions are given in the main text. Tables S5–S9 report incumbency by subgroup against the same permutation draws restricted to the subgroup. The nulls are N1 (market $\times$ year), N2 (adding buyer sector), and N3 (adding buyer province instead). All subgroup tests under N1 have $p < 0.001$, except for four very small groups: restricted procedure ($p = 0.24$, $n = 8$), Article 21(a) ($p = 0.04$, $n = 7$), and Article 21(d) and the year 2010 (one eligible contract each). Table S10 reports sensitivities; Figure S1 shows incumbency by year.

Further robustness checks give the following results. Excluding the 116 possible multi-lot tenders gives an observed incumbency of 0.455 (N1/N2/N3 nulls 0.062/0.109/0.209). Excluding natural-person winners gives 0.464 (0.066/0.116/0.214). The excess amounts to 1,919, 1,683, and 1,203 incumbent contracts under N1, N2, and N3.

Table S5: Incumbency by product market.

Product market	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health information systems	1,252	0.716	0.151 [0.136, 0.167]	0.157	0.407
Maintenance / support services	375	0.536	0.029 [0.013, 0.045]	0.139	0.122
Physical security / surveillance	182	0.418	0.030 [0.011, 0.055]	0.097	0.317
Other IT	112	0.393	0.019 [0.000, 0.045]	0.132	0.233
ERP / management software	745	0.392	0.027 [0.016, 0.039]	0.077	0.111
Network / data-centre infrastructure	328	0.384	0.023 [0.009, 0.040]	0.106	0.121
Smart city / traffic / OT	58	0.397	0.041 [0.000, 0.103]	0.104	0.314
Cybersecurity	161	0.398	0.052 [0.019, 0.087]	0.132	0.186
Computers / peripherals	281	0.327	0.012 [0.000, 0.025]	0.047	0.164
Software licences	602	0.334	0.039 [0.025, 0.055]	0.105	0.139
Education technology	80	0.350	0.068 [0.025, 0.125]	0.201	0.168
Call centre / help desk	19	0.316	0.062 [0.000, 0.158]	0.222	0.178
GIS / city information	230	0.326	0.088 [0.057, 0.122]	0.117	0.146
Custom software / web / mobile	488	0.201	0.006 [0.000, 0.014]	0.029	0.056

Table S6: Incumbency by buyer sector.

Buyer sector	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Health	1,502	0.667	0.129 [0.116, 0.142]	0.143	0.372
Other public	583	0.395	0.030 [0.017, 0.045]	0.075	0.120
Education	439	0.364	0.028 [0.014, 0.043]	0.118	0.164
Municipal/Local	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Provincial admin	115	0.374	0.046 [0.009, 0.087]	0.108	0.309
Social/Finance	196	0.357	0.041 [0.015, 0.066]	0.185	0.086
Agriculture/Environment	210	0.348	0.042 [0.019, 0.067]	0.110	0.100
Infrastructure/Transport	342	0.322	0.029 [0.015, 0.047]	0.097	0.116
Defence	131	0.298	0.039 [0.008, 0.076]	0.219	0.070
Security/Interior	100	0.280	0.025 [0.000, 0.060]	0.145	0.103
Justice	24	0.292	0.038 [0.000, 0.125]	0.292	0.091

S3.2 Nulls that preserve firm geography

A firm's home province is the province where it won most contracts. Null N4 permutes winners within market $\times$ year $\times$ winner's home province: the null mean is 0.185 (ratio 2.44, excess 1,313 contracts). Null N4b permutes within market $\times$ year $\times$ buyer province $\times$ winner's home province. In N4b, 31% of eligible contracts are in cells with a single possible winner. Its null mean is 0.242 (ratio 1.87, excess 1,034 contracts). All $p < 0.001$.

S3.3 Renewals

Titles are normalized by Turkish-aware lower-casing, removal of diacritics, digits, month names, and about 110 generic procurement words, and stemming to five characters; similarity is the Jaccard index of the stemmed word sets. A contract is a renewal if the buyer has a contract in the same market 183–548 days earlier with similarity of at least 0.5. In a check of 60 best-match pairs drawn at random within four similarity bands (15 each), coded from the titles with AI assistance (Claude), 38 of the 42 pairs above the threshold described the same recurring service. Nine of the 18 pairs below it did so as well, so the new-need group contains some renewals. Table S11 reports results by procedure and market and alternative definitions.

S3.4 Buyer dependence

Figure S2 shows the distribution of main-supplier shares. With per-buyer permutation p -values and Benjamini–Hochberg control of the false discovery rate at 5% across the 540 buyers, 273 buyers

Table S7: Incumbency by buyer type (keyword classes of buyer names).

Buyer type	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
MoH hospital / dental centre	617	0.831	0.177 [0.154, 0.203]	0.181	0.386
University hospital / medical faculty	110	0.627	0.099 [0.045, 0.155]	0.120	0.363
MoH provincial directorate / hospital union	695	0.563	0.098 [0.079, 0.118]	0.113	0.391
University (non-health unit)	401	0.362	0.025 [0.012, 0.040]	0.113	0.164
Municipality / municipal company	1,271	0.362	0.031 [0.023, 0.041]	0.057	0.143
Other	296	0.351	0.031 [0.014, 0.051]	0.086	0.183
Other health body (central/agency)	82	0.378	0.058 [0.012, 0.110]	0.139	0.130
Central government / agency	1,441	0.353	0.035 [0.027, 0.045]	0.124	0.107

Table S8: Incumbency by procedure.

Procedure	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
Article 21(b)	618	0.723	0.143 [0.120, 0.165]	0.159	0.376
Article 21(c)	34	0.412	0.020 [0.000, 0.088]	0.069	0.255
Open	3,041	0.441	0.057 [0.051, 0.064]	0.112	0.203
Article 21(f)	1,204	0.346	0.033 [0.023, 0.043]	0.080	0.131
Article 21(a)	7	0.286	0.049 [0.000, 0.286]	0.173	0.227
Restricted	8	0.125	0.033 [0.000, 0.125]	0.059	0.035
Article 21(d)	1	0.000	0.001 [0.000, 0.000]	0.000	0.000
Article 21(e)	0	–	– [–, –]	–	–

(50.6%) exceed their N1 null on the main-supplier share and 357 (66.1%) on the supplier HHI; under N2 the figures are 212 (39.3%) and 289 (53.5%). With 1,000 permutations, no buyer can pass a Bonferroni correction. The share of buyers above their 95th null percentile (uncorrected) rises with buyer size (50% for 5–9 contracts, 63% for 10–19, 80% for 20 or more). The mean excess, however, falls with size (0.24, 0.23, 0.13), so the rising share reflects the tighter null distributions of large buyers, that is, statistical power.

S4 Supplier choice

For each of the 4,913 repeat-eligible contracts, the choice set contains the winner and 30 firms sampled at random from those that won at least one contract in the same product market in the same or the previous year (149,670 rows). Table S12 reports the main specifications with buyer-clustered standard errors (1,126 buyers); a 500-draw buyer bootstrap gives an interval of 20.7–27.9 for the preferred incumbency odds ratio. If the home province is computed from all wins, including the contract being predicted, the incumbency odds ratio is 15.2 (12.9–17.8). Excluding the home-province variable gives 29.8. In a model with interactions, renewal multiplies the incumbency odds ratio by 3.7 (2.9–4.9) and Article 21(b) by 4.6 (3.2–6.6).

S5 Procedure and timing

S5.1 Contract-level model

The 21(b) $\times$ post-2018 odds ratio is 2.42 (1.43–4.09) with the buyer-history controls and two-way clustering, and 2.42 (1.57–3.73) with buyer clustering. Without the prior-contract and prior-supplier controls it is 2.90 (1.64–5.15); without any buyer-history controls it is 2.93 (1.65–5.19). The pre-2018 marginal effect of 21(b) is between –3.5 and –3.8 percentage points and insignificant in all specifications. Further checks give interaction odds ratios between 1.76 and 2.43: an IT-only sample, broad scope, recorded buyer labels, recorded firm names, a linear trend instead of year fixed effects, and excluding 2020. Alternative break dates give 2.55 (25 August 2017, KHK 694; log-likelihood

Table S9: Incumbency by year.

Year	Eligible	Observed	N1 null [95% interval]	N2 null	N3 null
2026	46	0.587	0.079 [0.022, 0.152]	0.151	0.301
2025	353	0.450	0.067 [0.045, 0.093]	0.112	0.222
2024	350	0.557	0.078 [0.054, 0.106]	0.144	0.263
2023	413	0.496	0.098 [0.075, 0.121]	0.131	0.265
2022	463	0.501	0.080 [0.058, 0.102]	0.132	0.260
2021	462	0.450	0.068 [0.050, 0.089]	0.106	0.222
2020	410	0.483	0.076 [0.054, 0.100]	0.131	0.224
2019	307	0.518	0.072 [0.046, 0.101]	0.122	0.262
2018	345	0.441	0.050 [0.029, 0.072]	0.095	0.147
2017	363	0.355	0.046 [0.028, 0.066]	0.082	0.148
2016	320	0.409	0.043 [0.025, 0.066]	0.087	0.174
2015	310	0.410	0.041 [0.023, 0.065]	0.098	0.167
2014	295	0.400	0.035 [0.017, 0.054]	0.091	0.165
2013	234	0.410	0.039 [0.017, 0.064]	0.084	0.148
2012	165	0.352	0.024 [0.006, 0.048]	0.060	0.137
2011	76	0.368	0.022 [0.000, 0.053]	0.118	0.136
2010	1	0.000	0.029 [0.000, 1.000]	0.033	0.000

Table S10: Sensitivity of the incumbency result (1,000 permutations each; all $p < 0.001$).

Variant	Observed	Null mean [95% interval]	Ratio
Main (N1: market $\times$ year)	0.452	0.062 [0.056, 0.067]	7.3
N1 with 3-year windows	0.452	0.060 [0.055, 0.066]	7.5
N2: market $\times$ year $\times$ sector	0.452	0.110 [0.103, 0.116]	4.1
Buyer = recorded label	0.462	0.106 [0.100, 0.112]	4.4
Firm = recorded name	0.394	0.047 [0.042, 0.052]	8.4
Excluding call-centre contracts	0.452	0.062 [0.056, 0.067]	7.3
Including ambiguous scope	0.435	0.056 [0.051, 0.061]	7.8
Cells = buyer sector $\times$ year	0.452	0.047 [0.042, 0.052]	9.7
Incumbency at buyer level (any market)	0.389	0.053 [0.049, 0.058]	7.3
Excluding Article 21(f) contracts	0.495	0.079 [0.071, 0.087]	6.3
N3: market $\times$ year $\times$ province	0.452	0.207 [0.201, 0.213]	2.2

−2801.1 against −2802.5 for 25 May 2018), 2.10 (placebo, May 2016, $p = 0.009$), and 1.57 (placebo, May 2020, $p = 0.08$). The interaction is 2.64 among health buyers and 0.86 ($p = 0.78$) among other buyers.

S5.2 Event study

Main-text Figure 4 reports the 21(b)–open gap by year relative to 2013–17. In that period the gap does not differ across years (joint tests $p = 0.19$ for all buyers and 0.20 for health buyers). In 2018 it is 37 percentage points higher for all buyers (24–50) and 43 points higher for health buyers (23–62). From 2019 on it averages 14 points above the reference for all buyers ($p = 0.003$) and 12 points for health buyers ($p = 0.04$). With buyer fixed effects, health buyers show a gap averaging −10 points in 2013–17, positive in every year from 2018, and 24 points (10–38) above the reference from 2019 on. Among parametric shapes, a step in August 2017 combined with a declining trend fits best. A step in May 2018 fits worse by 3.4–7.8 AIC points across the three samples, and a pure trend fits about as badly as no change. Only 8 and 10 repeat-eligible 21(b) contracts fall in the windows 25 August–31 December 2017 and 1 January–24 May 2018. The years 2010–12, with little buyer history and 24 21(b) contracts, show a high gap as well. Against that base, no later year is significantly higher.

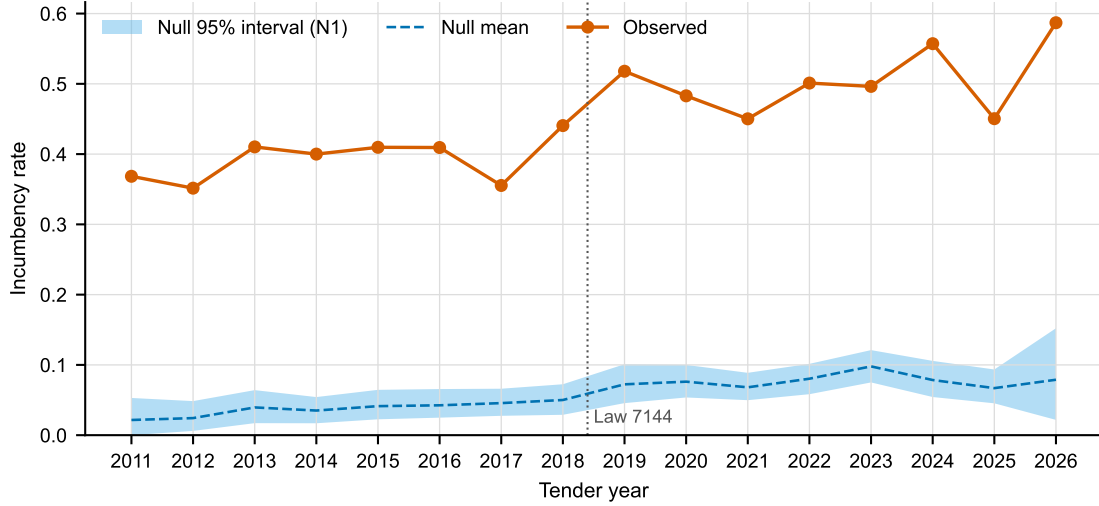


Figure S1: Incumbency rate by year, observed and under null N1 (mean and 95% interval).

Table S11: Incumbency of renewals and new needs: observed [N1 null; N3 null].

	Renewals	New needs
All	0.720 [0.092; 0.288] ($n=1,743$)	0.305 [0.045; 0.163] ($n=3,170$)
Article 21(b)	0.900 [0.198; 0.427] (291)	0.566 [0.093; 0.331] (327)
Open	0.669 [0.077; 0.267] (1,128)	0.307 [0.045; 0.165] (1,913)
Other (mainly 21(f))	0.735 [0.049; 0.237] (324)	0.210 [0.026; 0.099] (930)
Health information systems	0.869 [0.209; 0.452] (519)	0.607 [0.110; 0.375] (733)
ERP / management software	0.821 [0.034; 0.227] (201)	0.233 [0.024; 0.069] (544)
Software licences	0.593 [0.048; 0.212] (241)	0.161 [0.034; 0.091] (361)
Custom software	0.544 [0.010; 0.152] (79)	0.134 [0.006; 0.038] (409)
Maintenance / support	0.776 [0.036; 0.138] (192)	0.284 [0.021; 0.107] (183)
Network / data centre	0.670 [0.035; 0.184] (112)	0.236 [0.016; 0.090] (216)
Computers / peripherals	0.436 [0.014; 0.194] (110)	0.257 [0.010; 0.146] (171)
GIS / city information	0.828 [0.135; 0.274] (58)	0.157 [0.071; 0.104] (172)
Threshold 0.40	0.709 [-; 0.286]	0.296 [-; 0.159]
Threshold 0.67	0.744 [-; 0.302]	0.329 [-; 0.168]
Window 0–18 months	0.695 [-; 0.293]	0.268 [-; 0.143]
Window 0–24 months	0.684 [-; 0.290]	0.250 [-; 0.135]

S5.3 Dyad continuation

For each new buyer–firm pair in a product market, we ask whether the pair signs another contract in that market. Of 7,761 pairs, 4,334 have no later contract by the buyer in that market and cannot continue. The analysis uses the remaining 3,427, of which 34.6% continue. Covariates are measured at the first contract without look-ahead. The number of later opportunities (the buyer’s later contracts in the market) enters as a covariate, because the offset restriction is rejected (coefficient 0.90 in Poisson pseudo-maximum likelihood, $p = 0.02$; 0.88 in NB2, $p = 0.006$). In NB2 the dispersion parameter is estimated ($\alpha = 0.46$, SE 0.12; likelihood-ratio statistic for $\alpha = 0$: 152.5). Firm size and prior distinct buyers are highly collinear ($r = 0.99$; VIFs of 63 and 61), and generalism correlates 0.86 with size, so these variables are not separately identified in the full model. Entered alone, generalism and size are both positively associated with continuation. There is no evidence that specialist firms are more likely to keep a buyer. A first contract under 21(b) is associated with continuation (odds ratio 1.85).

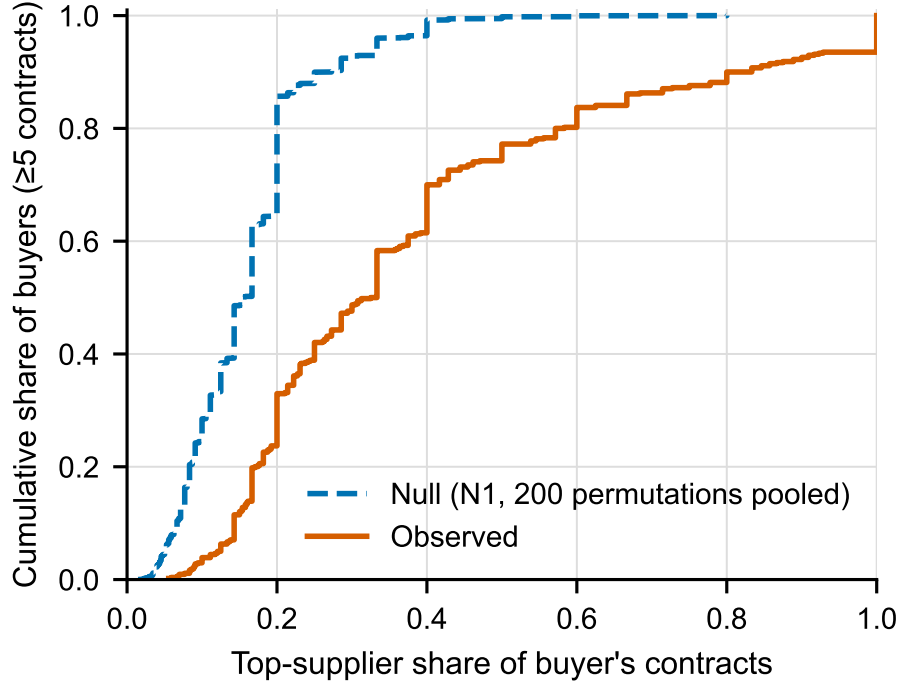


Figure S2: Distribution of the main supplier's share of each buyer's contracts (540 buyers with at least five contracts), observed and under null N1.

Table S12: Conditional logit: incumbency odds ratio [buyer-clustered 95% CI] by specification and subgroup.

Specification / subgroup	Contracts	Incumbent OR
Home province from prior wins (preferred)	4,913	23.8 [20.6, 27.5]
Renewals	1,743	56.7 [44.7, 72.0]
New needs	3,170	13.9 [11.7, 16.6]
Article 21(b)	618	68.0 [50.5, 91.7]
Open	3,041	19.5 [16.5, 23.0]
Home province from all wins	4,913	15.2 [12.9, 17.8]
Renewals / new needs		32.7 / 8.8
Article 21(b) / open / other		52.5 / 12.8 / 10.5
Without home province	4,913	29.8 [26.0, 34.1]

S6 Single bidding

Random sample. Bid counts were retrieved from published result announcements for a random sample of tenders stratified by year (40 per year drawn from all retrieved tenders). Of these, 393 are in the main sample and report bid counts; none is available for 2010–2011. Rates are weighted by the inverse sampling fraction of each year, with linearized 95% intervals. Incumbency status is undefined when the tender is the buyer's first in the market. Adjusting for procedure and year (logit, buyer-clustered, $n = 208$ tenders with defined incumbency), the odds ratio of a single bid for incumbent winners is 4.44 (2.39–8.25). The single-bid rate has a weak upward trend (odds ratio 1.05 per year; $p = 0.04$ unweighted, $p = 0.11$ weighted). In 308 tenders with a recorded number of firms that obtained the documents, design-weighted averages are 4.0 such firms and 1.9 valid bids. In 45% of these tenders two or more firms obtained the documents but only one valid bid was received.

Software sample. An interrupted enrichment run of the scraper retrieved result details for 3,400 tenders found with the first search keyword, *yazılım* (software). For tenders from 2021 on, these include the number of valid bids. Joined to the main sample, 1,194 tenders have this count,

Table S13: Dyad continuation, logit (full specification; product-market and year fixed effects; two-way clustered 95% CIs).

	OR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.85 [1.37, 2.50]	<0.001
First contract other negotiated (vs open)	1.26 [1.03, 1.54]	0.025
Log real value of first contract	1.02 [0.95, 1.10]	0.552
Log firm prior contracts (other buyers)	1.58 [0.97, 2.58]	0.067
Log firm prior markets (generalism)	1.20 [0.86, 1.66]	0.284
Log firm prior distinct buyers	0.69 [0.41, 1.15]	0.156
Log buyer prior contracts	0.76 [0.69, 0.85]	<0.001
Log later opportunities of buyer in market	1.89 [1.68, 2.13]	<0.001

Table S14: Number of later contracts, NB2 with estimated dispersion (full specification).

	IRR [95% CI]	<i>p</i>
First contract under 21(b) (vs open)	1.61 [1.39, 1.87]	<0.001
First contract other negotiated (vs open)	1.19 [1.02, 1.39]	0.029
Log real value of first contract	1.01 [0.96, 1.06]	0.691
Log firm prior contracts (other buyers)	1.36 [1.07, 1.74]	0.012
Log firm prior markets (generalism)	1.20 [0.98, 1.45]	0.074
Log firm prior distinct buyers	0.76 [0.59, 0.99]	0.045
Log buyer prior contracts	0.77 [0.72, 0.82]	<0.001
Log later opportunities of buyer in market	2.42 [2.23, 2.63]	<0.001

about 94% of software-titled tenders from 2021 on but only 35% of all tenders in that period. The sample is therefore not random. Single valid bids occur in 66.7% of incumbent-won tenders and 49.3% of others. In a logit with procedure, year, and market fixed effects (681 tenders with defined incumbency, buyer-clustered), the incumbent odds ratio is 2.04 (1.40–2.96), and 2.65 when renewal status is added.

S7 Network structure

Graph. The main-sample graph has 2,814 firms, 2,890 buyers and 7,219 weighted edges, with density 8.9×10^{-4} . The giant component contains 4,446 nodes (78%) and 92% of contracts. Degree distributions (number of distinct partners) are heavy-tailed. Fitted by maximum likelihood with the protocol of Clauset, Shalizi and Newman (2009), firms have $\hat{\alpha} = 2.21$, and a lognormal is not distinguishable from a power law ($R = -1.67$, $p = 0.09$). For buyers ($\hat{\alpha} = 2.06$) the power law is rejected by the bootstrap goodness-of-fit test, and a lognormal fits better ($R = -6.19$, $p < 0.001$). Degree assortativity is -0.17 against -0.04 in the null. Bipartite clustering exceeds the null: Robins–Alexander 0.026 against 0.012, Latapy 0.31 against 0.25. Figure S3 shows the core of the network.

Communities. Over 50 Louvain runs with random seeds and node orders, weighted modularity of the giant component ranges from 0.760 to 0.764 (34–43 communities). Barber’s bipartite modularity of the same partitions is practically identical. In 200 bipartite degree-preserving randomizations (curveball), the unweighted modularity of the giant component is 0.694 ± 0.003 against an observed 0.731, and the weighted modularity is 0.756 ± 0.003 against an observed 0.761. Communities align with buyer sectors (NMI 0.25; permuted labels 0.03; purity 0.64 against a largest-label baseline of 0.44), and ten communities consist of at least 91% health buyers. They align only weakly with firms’ main product markets (NMI 0.13; permuted 0.04). In annual full graphs, modularity exceeds its bipartite null only modestly (annual z between 0.4 and 3.9). Within annual giant components it does not exceed the null in 16 of 17 years, so we do not interpret yearly modularity as evidence of persistent segmentation.

Projections and brokerage. We computed exact Poisson-binomial p -values for co-occurrences

Table S15: Single-bid rates (design-weighted random sample).

Group	Tenders	Single bid, % [95% CI]	Single valid bid, %
All tenders	393	44.2 [39, 49]	55.8
Open	225	40.8 [34, 48]	55.9
Article 21(b)	50	75.0 [60, 86]	81.6
Article 21(f)	114	40.0 [31, 50]	46.9
Other negotiated	4	0.0	18.5
Winner is incumbent	95	71.2 [61, 80]	76.7
Winner not previous supplier in market	113	32.4 [24, 42]	48.4
Buyer's first contract in market	185	38.3 [31, 46]	50.2
Before 25 May 2018	140	38.6 [31, 47]	49.1
After 25 May 2018	253	48.9 [42, 55]	61.5

under the bipartite configuration model (BiCM). With a false-discovery-rate correction over all pairs at 0.05 or 0.01, no link is validated in either projection, because the data are sparse: 94–97% of co-occurring pairs share exactly one neighbour. A liberal variant that corrects only over co-occurring pairs retains 15,473 firm links and 42,005 buyer links. It essentially reproduces the raw projection without hub-mediated links, so we do not treat it as a validated backbone. Disparity-filter backbones of Newman-weighted projections at $\alpha = 0.05$ contain 1,077 firm links (modularity 0.73) and 2,685 buyer links (0.73). They overlap little with the liberal BiCM set (Jaccard 0.04).

Across these projections and the raw projection, firm betweenness correlates with the number of product markets a firm serves, but also with its number of buyers. Controlling for buyers and contracts, the partial Spearman correlation is 0.12–0.15, and adding the number of markets to a size-only regression raises R^2 by 0.003–0.015. Brokerage thus adds little beyond firm size. In the raw buyer projection, the most connected buyer, a provincial health directorate once pooled labels are split, shares at least one supplier with 19% of the projection's giant component. With recorded pooled labels the figure is 45%, which shows how strongly such statistics depend on buyer coding.

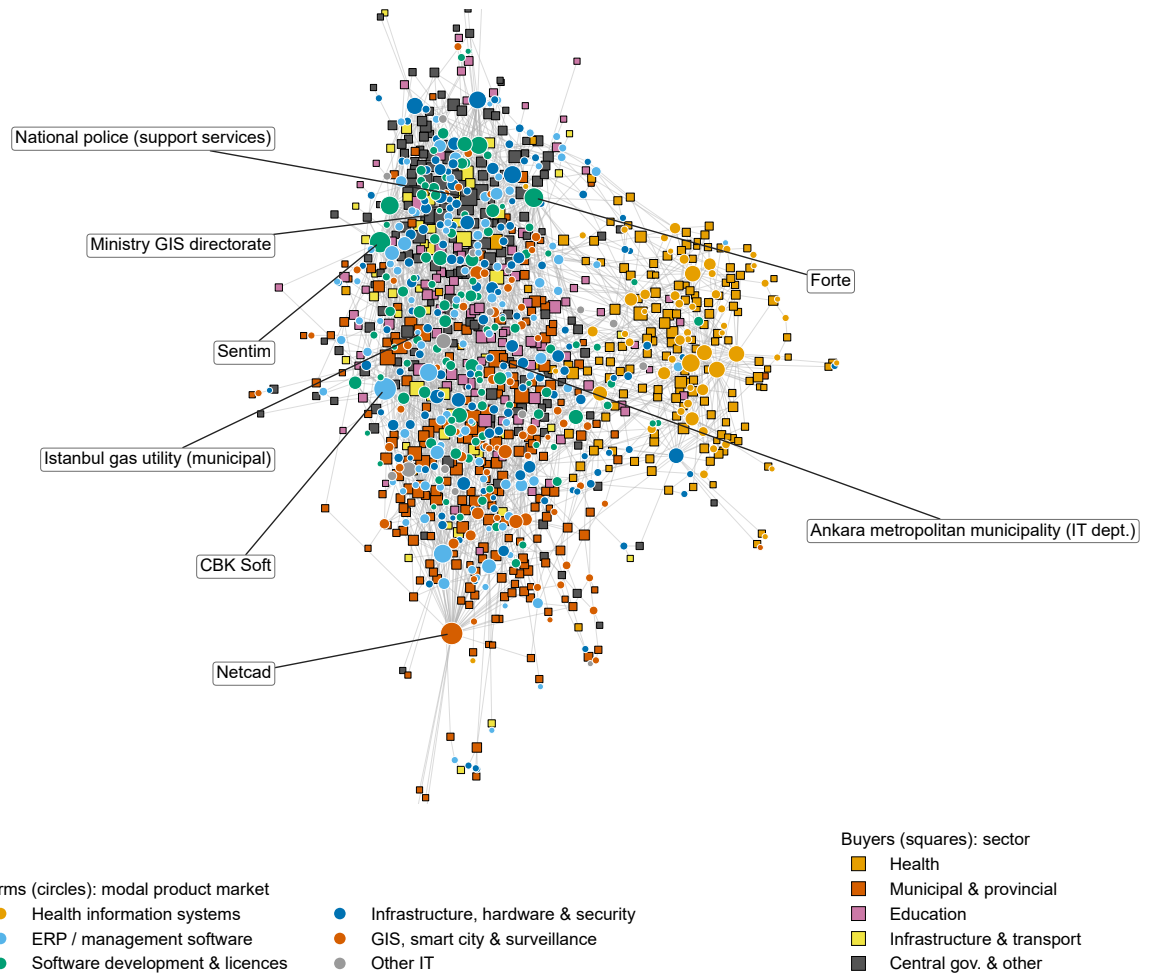


Figure S3: Core of the buyer-supplier network (nodes of degree ≥ 3 , largest connected component; force-directed layout with fixed seed). Circles are firms coloured by their main product market; squares are buyers coloured by sector.